Shirin Rajabpour

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Research Interests

Deep learning for medical imaging and clinical AI; neuroimaging and developmental brain mapping; graph analysis of structural brain connectivity; biosignal analysis (EEG/MEG) and source modeling; pattern recognition and machine learning for healthcare data.

Education

K. N. Toosi University of Technology **Oct 2020 – Sep 2024**
M.Sc. in Biomedical Engineering – Bioelectric | GPA: 17.75/20 *Tehran, Iran*

- Thesis: *Modeling Neonatal Development of Perisylvian Regions Using Graph Analysis of Structural Covariance Networks*
- Supervisor: Dr. Hamid Abrishami Moghaddam

Yazd University **Sep 2015 – Sep 2019**
B.Sc. in Electrical Engineering – Electronics | Final-year GPA: 16.37/20 *Yazd, Iran*

- Thesis: *Robust nonlinear control schemes for finite-time tracking of a 5-DOF robotic exoskeleton*
- Supervisor: Dr. Ali Abooee

Research Experience

Research Assistant **Oct 2020 – Sep 2024**
Machine Vision & Medical Image Processing Laboratory, K. N. Toosi University of Technology *Tehran, Iran*

- Conducted M.Sc. thesis research on neonatal perisylvian development using graph analysis of structural covariance networks from infant structural MRI (ages 0–24 months), with NetworkX and related neuroinformatics tools
- Developed and evaluated machine learning and deep learning approaches for medical image analysis in a neuroimaging laboratory setting
- Collaborated on interdisciplinary projects spanning medical imaging, neuroinformatics, and machine learning; contributed to manuscript preparation

Publications

Manuscripts in preparation (not peer-reviewed):

- **Rajabpour, S.**, Moghaddam, H.A. "Modeling the Neonatal Development of Perisylvian Regions Using Graph Analysis of Structural Covariance Networks." Manuscript in preparation.
- **Rajabpour, S.** et al. "Tumor Mask and Masked CT Fusion for 3D CNN Classification of Kidney Tumors." Manuscript in preparation.

Selected Projects

Infant Brain Development Analysis | *Python, PyTorch, NetworkX, Medical Imaging* **Jan 2023 – Jan 2025**

- Developed a graph analysis framework for structural covariance networks in early brain maturation (200+ infant MRI scans); applied ML methods to neuroimaging-based developmental analysis

EEG/MEG Source Analysis | *MATLAB, Brainstorm* **Mar 2022 – Jul 2022**

- Analyzed electrophysiology data in Brainstorm for source localization; implemented and evaluated forward and inverse EEG/MEG methods (Functional Brain Imaging Systems, Dr. Ali Khadem)

EEG Signal Classification | *Python, MATLAB, Signal Processing* **Jan 2022 – Jul 2022**

- Entropy-based feature extraction in the EMD–DWT domain for focal versus non-focal EEG discrimination on 10,000+ segments (Functional Brain Imaging Systems, Dr. Ali Khadem)

Breast Cancer Prediction Models | *Python, SVM, Neural Networks* **Jan 2021 – Jan 2022**

- SVM ensembles and Bayes MLE classification (Wisconsin dataset); neural network models for breast cancer detection (Pattern Recognition, Dr. H. A. Moghaddam; Neural Networks, Dr. M. Teshnehlab)

Teaching Experience

Teaching Assistant | Medical Imaging Systems **Sep 2023 – Jan 2025**
Master's course taught by Dr. Ali Khadem, K. N. Toosi University of Technology *Tehran, Iran*

- Assisted students with homework; organized lecture notes; graded assignments; held student meetings; designed exam questions

Teaching Assistant | Digital Image Processing **Feb 2023 – Jun 2023**
Graduate course taught by Dr. H. A. Moghaddam, K. N. Toosi University of Technology *Tehran, Iran*

- Graded homework assignments

Teaching Assistant | Introduction to Electrical Engineering **Feb 2023 – Jun 2023**
Undergraduate course taught by Dr. H. A. Moghaddam, K. N. Toosi University of Technology *Tehran, Iran*

- Graded examination papers

Honors & Awards

- **Best Seminar Award** in Medical Engineering, Faculty of Electrical Engineering, K. N. Toosi University of Technology (Oct 2023)
- Ranked in the **top 2%** of ∼14,000 candidates in the national M.Sc. entrance examination, Electrical Engineering (Oct 2020)
- Ranked in the **top 2%** of ∼182,000 candidates in the national B.Sc. entrance examination (Sep 2015)

Professional Experience

- AI-DAY Committee Member, K. N. Toosi University of Technology (Jan 2022 – Jan 2023)
- IEEE Educational Activities Committee Member, K. N. Toosi University of Technology (Jan 2020 – Jan 2023)

Technical Skills

Programming: Python, MATLAB, SQL, C++, C#
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, LLM
Imaging & Biosignals: OpenCV, SciPy, EEGLAB, Brainstorm
Neuroinformatics: NetworkX, GRETNA, GraphVar
Other: Power BI, NumPy, Pandas, Matplotlib, Seaborn, Git, LaTeX, Linux

Selected Coursework

- Pattern Recognition
- Digital Image Processing
- Digital Signal Processing
- Functional Brain Imaging Systems
- Neural Networks
- Biomedical Instruments
- Medical Imaging Systems
- Seminar

Certifications

- **Deep Learning with TensorFlow** – May 2021
- **Introduction to Machine Learning** – September 2020
- **Python Programming** – September 2020
- **Advanced MATLAB** – July 2017

Languages

- **English**: Fluent (TOEFL iBT 100/120)
- **Persian**: Native

References available upon request.